
Stochastic Processes and Quantum Theory

Based on arXiv:2302.10778

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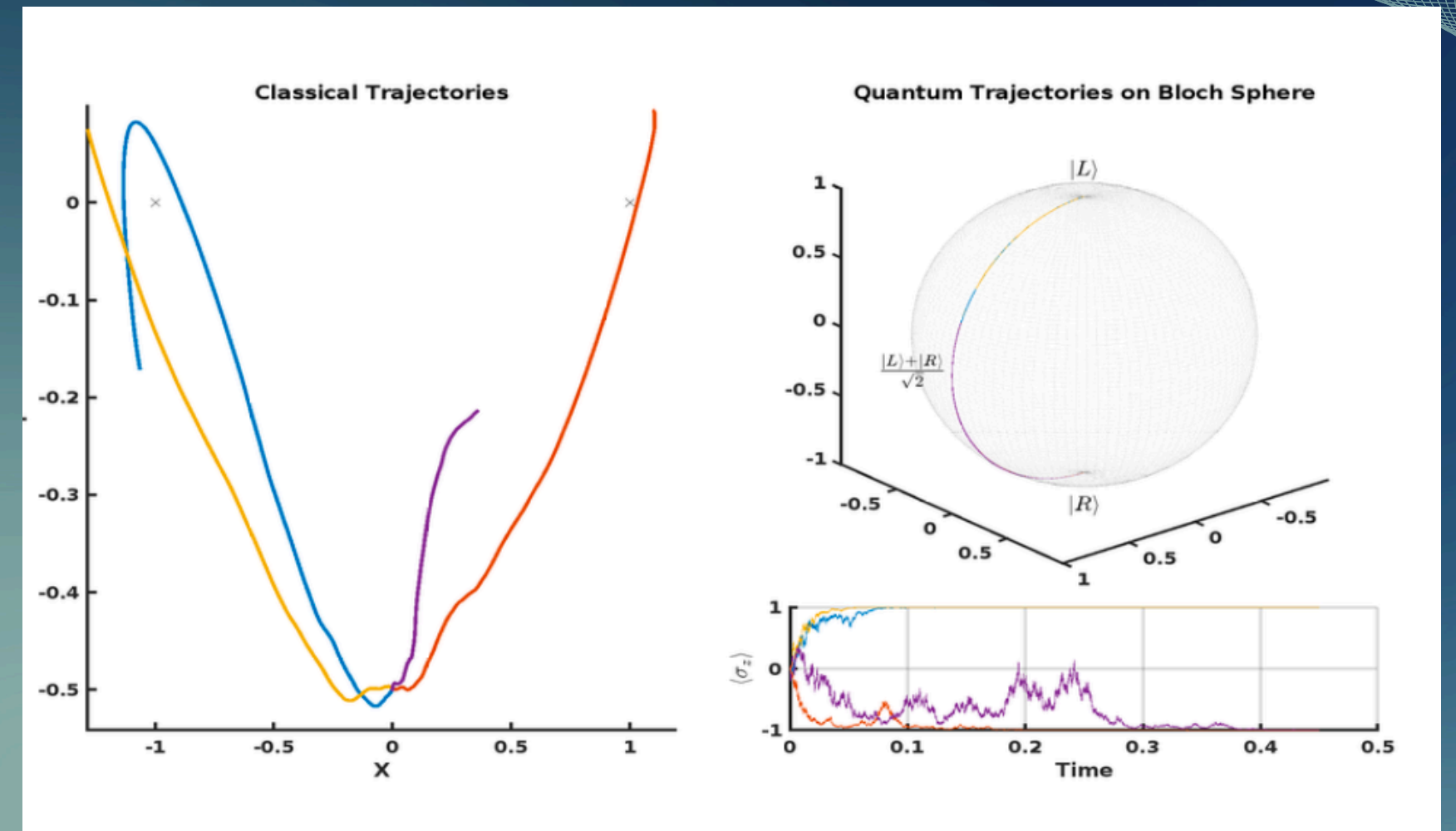
SUMMER 2025

Motivation

- Theoretical aspects
Failures of quantum theory
- Practical aspects
Simulations of quantum systems
Complex stochastic systems analysis

View 1: approximate dynamics

- Rooted in quantum:
Oppenheim et al. (2023)
- Rooted in stochastic processes:
Path integral methods,
Vittorio Visco et al. (2025)

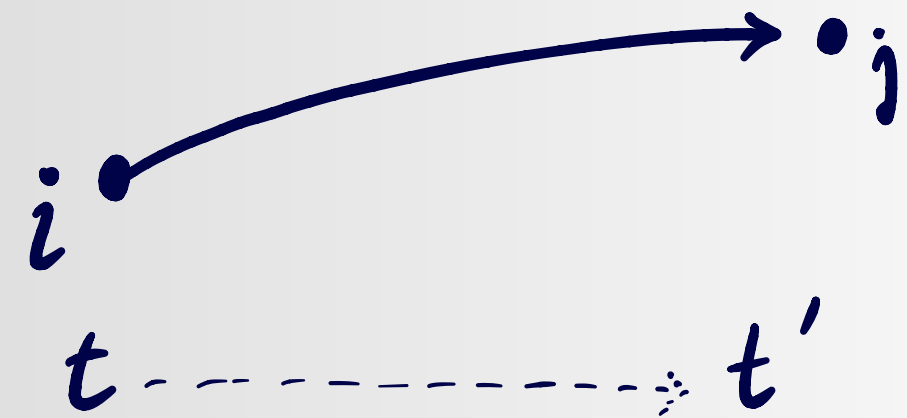
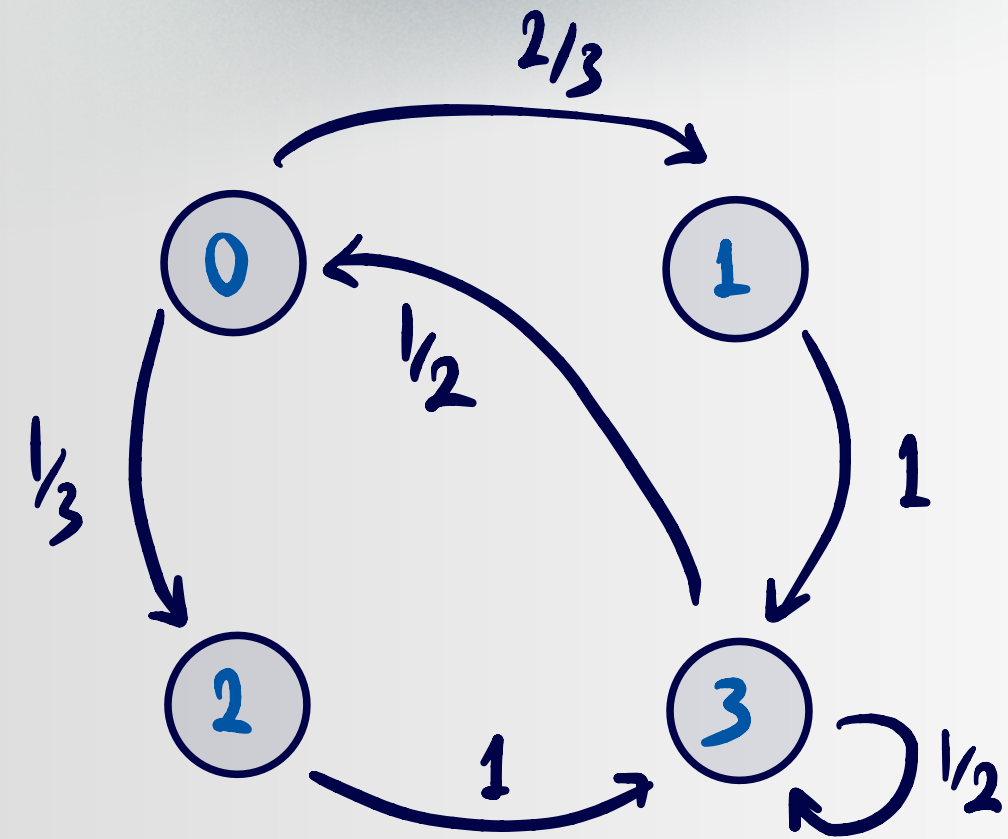


View 2: exact correspondence

Transition matrices

$$\Gamma(n+1 \leftarrow n) = \begin{pmatrix} 0 & \frac{2}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \end{pmatrix}$$

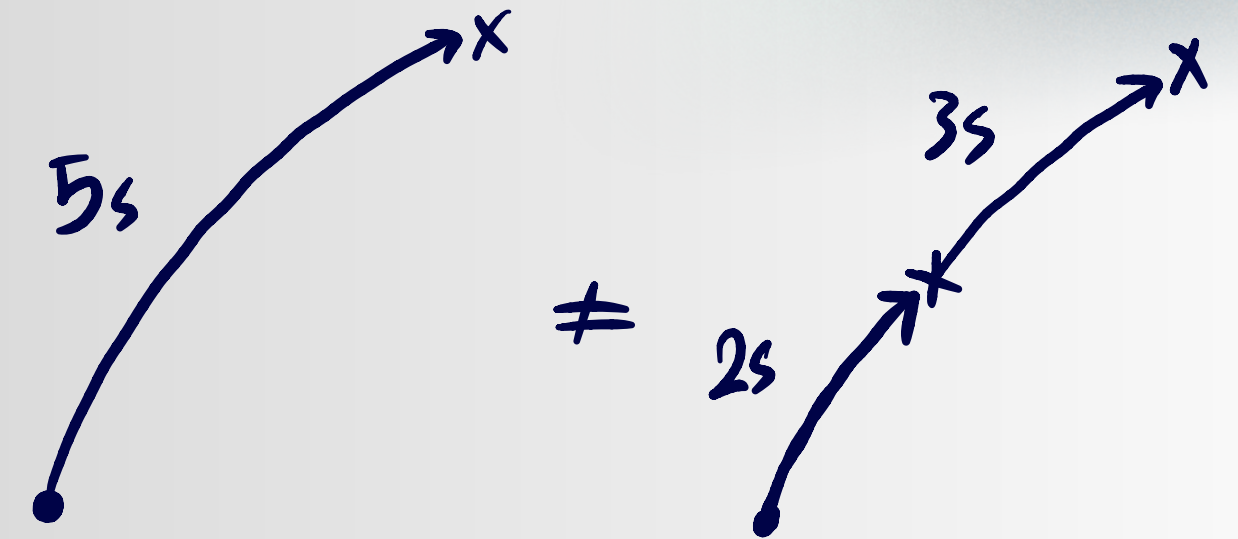
$$\Gamma_{ij}(t \leftarrow t_0) \equiv p(i, t | j, t_0),$$



INDIVISIBILITY AND NON-MARKOVIAN

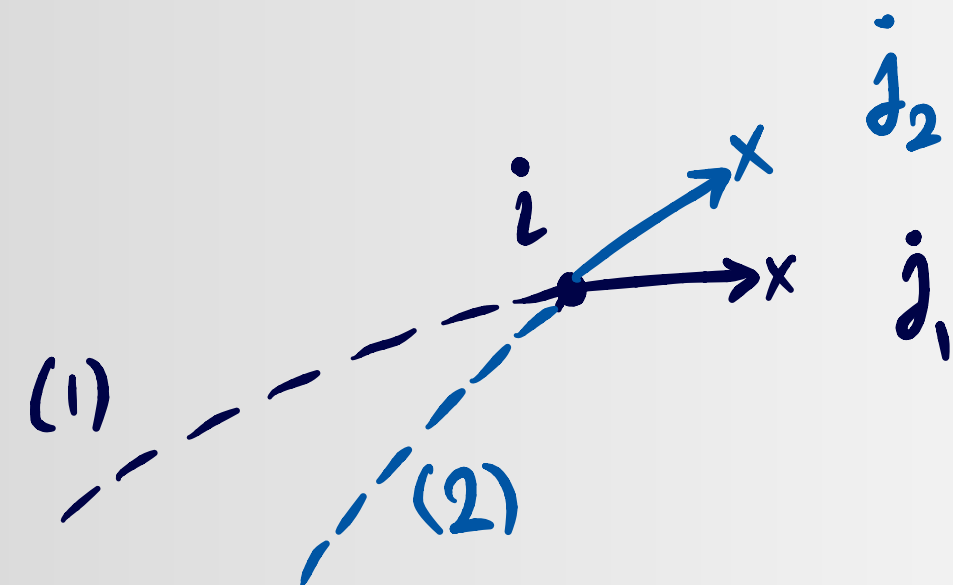
Indivisibility:

$$\Gamma(t \leftarrow t_0) \neq \Gamma(t \leftarrow t')\Gamma(t' \leftarrow t_0)$$



non-markovian

$$\Gamma(t_1 + \Delta t \leftarrow t_1) \neq \Gamma(t_2 + \Delta t \leftarrow t_2)$$



STOCHASTIC-QUANTUM DICTIONARY

$$\Gamma_{ij}(t \leftarrow 0) = |\Theta_{ij}(t \leftarrow 0)|^2$$

$$\Psi(t) \equiv \Theta(t \leftarrow 0)e_j$$

$$p_i(t) = |\Psi_i(t)|^2$$

$$\langle A(t) \rangle = \Psi^\dagger(t) A(t) \Psi(t)$$

STOCHASTIC-QUANTUM DICTIONARY

Emeding: $N \rightarrow N' \leq N^3$

$$\Gamma_{ij}(t \leftarrow 0) = |U_{ij}(t \leftarrow 0)|^2$$

$$\Gamma_{ij}(t \leftarrow 0) = \text{tr}(U^\dagger(t \leftarrow 0)P_i U(t \leftarrow 0)P_j)$$

With $U_{(t \leftarrow 0)}^\dagger = U_{(t \leftarrow 0)}^{-1}$

STOCHASTIC-QUANTUM DICTIONARY

$$H(t) \equiv i\hbar \frac{\partial U(t \leftarrow 0)}{\partial t} U^\dagger(t \leftarrow 0)$$

Hamiltonian

$$i\hbar \frac{\partial \Psi(t)}{\partial t} = H(t) \Psi(t)$$

Schrödinger equation

$$\frac{d\langle A(t) \rangle}{dt} = \frac{i}{\hbar} \text{tr}([H(t), A(t)] \rho(t)) + \left\langle \frac{\partial A(t)}{\partial t} \right\rangle \langle A(t) \rangle$$

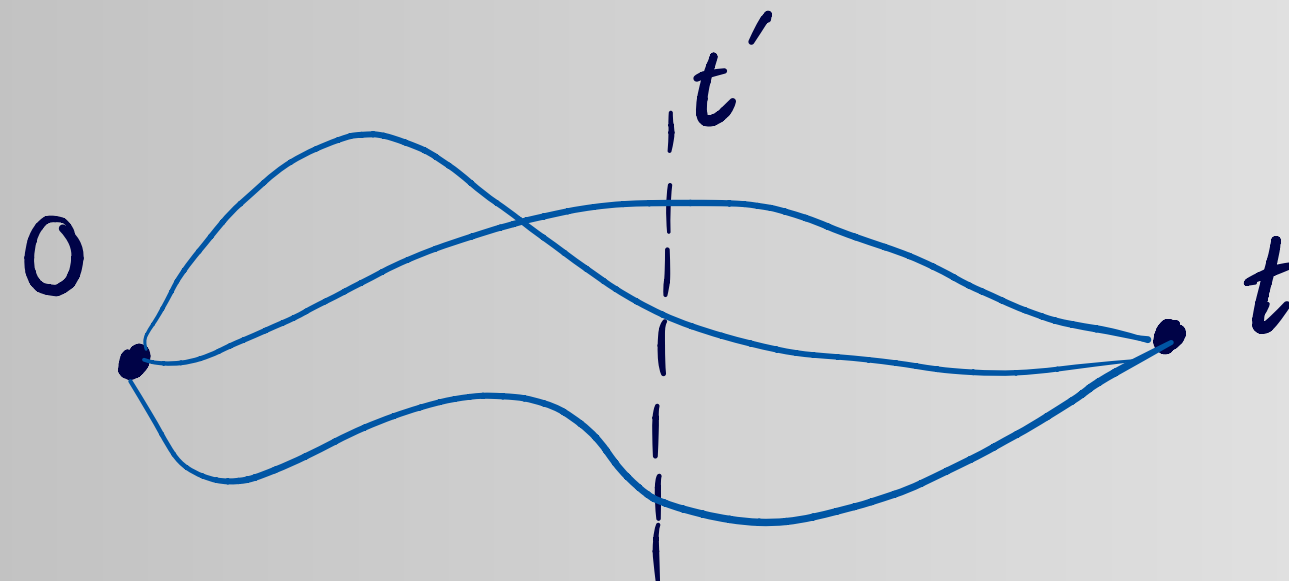
Ehrenfest
equation

SUPERPOSITION AND INDIVISIBILITY

$$\Gamma_{ij}(t \leftarrow 0) - [\Gamma(t \leftarrow t')\Gamma(t' \leftarrow 0)]_{ij} = \sum_{k \neq l} (U_{ik}(t \leftarrow t')\Psi_k(t'))^\dagger U_{il}(t \leftarrow t')\Psi_l(t')$$

$$= \sum_{k \neq l} (U_{ik}(t \leftarrow t')\Psi_k(t'))^\dagger U_{il}(t \leftarrow t')\Psi_l(t')$$

$$= \sum_{k \neq l} (\langle i | U_{(t \leftarrow t')} | k \rangle \langle k | U_{(t' \leftarrow 0)} | j \rangle)^\dagger (\langle i | U_{(t \leftarrow t')} | l \rangle \langle l | U_{(t' \leftarrow 0)} | j \rangle)$$



QUESTIONS TO THINK ABOUT

- How can non-trivial classical systems be investigated using the definition of indivisibles?
- what is the relation between indivisible and non markovian?

THANK YOU
